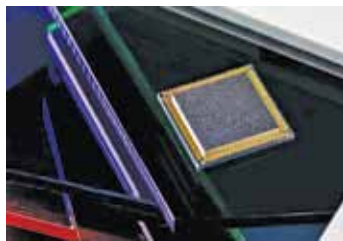




transparent sensor that detects both facial muscle deformation and vocal cord vibrations. This innovation, coupled with machine learning algorithms, enables precise emotion recognition, even with masked individuals. The system has been successfully implemented in a virtual reality digital concierge application. It operates on friction charging, eliminating the need for external power sources. The team's semi-curing technique for transparent conductors and a personalised mask created through multi-angle shooting enhance the system's customisation and effectiveness.

High-performance quantum computer with tunable-coupler concept

IQM Quantum Computers has advanced its quantum computing technology with a 20-qubit quantum computer that incorporates quantum processing units (QPUs) based on the tunable-coupler concept. This design achieves fast



(Credit: <https://thequantuminsider.com>)

two-qubit gate speed and high fidelity, with a median two-qubit gate fidelity of 99.51 % across 30 qubit pairs and a peak fidelity of 99.8 %. The system's benchmarks include a quantum volume (QV) of 32, measuring both scale and qual-

ity, and a circuit layer operations per second (CLOPS) of 2600, indicating throughput speed. Additionally, the system achieved a 20-qubit GHZ state with fidelity greater than 0.5, demonstrating multiqubit entanglement, and a Q-score of 11, showing its ability to solve the max-cut combinatorial optimisation task. IQM is currently working with prototype systems up to 150 qubits, aiming to provide high-performance systems for end-users.

Flexible, high-efficiency solar cells for vertical installations

Scientists at Nagoya University in Japan have developed a new material, fullerene indanone-based (FIDO), to enhance



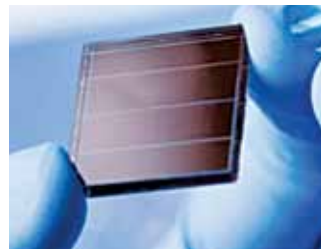
Amorphous solar cells with FIDO technology are more efficient, stable, and lightweight (Credit: Yutaka Matsuo)

the durability of next-generation perovskite solar cells. These cells are known for their high efficiency and flexibility, making them suitable for vertical installations like windows and walls. Fullerenes, carbon-based semiconductors with a

unique shape, play a crucial role in these cells by channeling electrons to generate power. The team used FIDO to create an amorphous solar cell film, which differs from the crystalline structure of traditional silicon solar cells. This amorphous material is more efficient and stable than standard films, and its properties do not deteriorate over time. The researchers believe their fullerene derivatives could have broader applications in organic electronics, including organic photodiodes and photodetectors, which could improve the resolution of imaging sensors in cameras and enable fingerprint authentication on smartphone displays.

Highly efficient solar cells designed with a thin coating of tetracene

Physicists at Paderborn University have developed a highly efficient solar cell design by coating silicon solar cells with a thin layer of the organic compound tetracene, us-



Physicists at Paderborn University have enhanced solar cell efficiency significantly using tetracene, an organic material, based on complex computer simulations. They discovered that defects at the tetracene-silicon interface boost energy transfer, promising a new solar cell design with drastically improved performance

ing advanced computer simulations. This innovative approach, detailed in Physical Review Letters, addresses the efficiency limits of silicon solar cells, which often waste energy as heat. Professor Wolf Gero Schmidt highlights the vast potential of solar energy, explaining that the tetracene layer converts short-wave light into high-energy excitations, increasing electricity generation.

The research, conducted with Dr Marvin Krenz and Prof. Dr Uwe Gerstmann, found that specific defects at the interface between tetracene and silicon accelerate exciton transfer, crucial for improved efficiency. These findings pave the way for a new type of solar cell with significantly enhanced performance, contributing to cleaner and more renewable energy generation.

New 3D printer could drastically reduce manufacturing costs and waste

MIT researchers have developed a 3D printing technique that enables the creation of high-performance solenoids, key components in electronics, using a custom 3D printer. This innovation could drastically reduce manufacturing costs and waste, and potentially make vital equipment like dialysis machines more accessible in remote or resource-limited areas. The printer is modified to use multiple mate-